

Sagar Koirala

Embedded Systems Engineer

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WORK EXPERIENCE

[D.T.C. Enterprise Public Company Limited](#)

Apr. 2025 - Jun. 2025

Embedded Systems Intern

Bangkok, Thailand

- **CAN Diagnostics:** Developed CANSense, a modular C++ and Qt tool for vehicle CAN bus analysis, reducing manual reverse-engineering and signal decoding time by 70% through real-time pattern-based highlighting of changing signal IDs.
- **STM32 Firmware:** Coded STM32 firmware to capture and stream live vehicle CAN logs over RS232 to a diagnostic tool, replacing slow Excel-based manual decoding with real-time pattern recognition.

[International Institute of Space Technology](#)

Feb. 2023 - Aug. 2024

Research Assistant - CubeSat ADCS System

Bangkok, Thailand

- **OTA Firmware:** Programmed a fault-tolerant secondary bootloader for STM32 microcontrollers using custom CAN communication protocols, flash memory protection, and CRC verification to execute reliable over-the-air firmware updates for satellite subsystems.
- **PCB Design:** Designed STM32F1-based PCBs for a custom magnetic torquer testing rig using Altium Designer, integrating magnetometers, current sensors, and motor drivers to measure physical output magnetic field strength accurately.

[Antarikhya Pratisthan Nepal](#)

Mar. 2021 - Jun. 2022

Satellite Research Fellow

Kathmandu, Nepal

- **Ground Station Operations:** Operated UHF radio equipment and tracking software to receive telemetry data during the orbital passes of the BIRDS-3 CubeSat.
- **LoRa Sensor Terminal Prototyping:** Developed remote LoRa-based telemetry terminals for glacial lake monitoring, integrating GSM communication backups and rugged environmental packaging to guarantee continuous data transmission under extreme cold and moisture constraints.

KEY ENGINEERING PROJECT

[Cellular IoT Remote Telemetry System](#)

Jul. 2025 - Apr. 2026

- **4-Layer PCB:** Designed a 100x70mm 4-layer PCB in Altium Designer featuring an ARM Cortex-M4 STM32 microcontroller, SIM7600G-H LTE/GNSS modem, CAN transceiver with TVS ESD protection, and a regulated automotive power tree architecture.
- **RTOS Firmware:** Developed FreeRTOS firmware implementing five tasks with zero heap allocation, streaming raw vehicle CAN data packets over UDP wireless links via a SIM7600 modem to a remote server.

[TelloESP32 Arduino Library Drone Telemetry](#)

Nov. 2024 - Jan. 2025

- **C++ Drone Library:** Authored an open-source C++ Arduino library enabling direct UDP socket control of DJI Tello drones from ESP32 microcontrollers, supporting real-time IMU telemetry parsing and dual-core firmware task scheduling.
- **Video Streaming:** Integrated H.264 video decoding callbacks to process drone video feeds, implementing multi-threaded network sockets to receive raw byte streams and transmit command signals with under 100ms system latency.

SKILLS

- **Hardware Design:** Altium Designer, KiCad, Schematic Capture & PCB Layout (4-layer), Analog/Digital Circuit Design, DFM, Component Sourcing, Gerber Generation
- **Firmware & Embedded:** C/C++, FreeRTOS, STM32 HAL, ESP-IDF, Bare-Metal / Bootloader Design, OTA Updates
- **Protocols & Interfaces:** CAN/CAN-FD, UART, SPI, I²C, USB 2.0(HID, CDC), TCP/UDP, Wi-Fi, LTE
- **Hardware Debugging:** Oscilloscope, Logic Analyzer, Multimeter, JTAG/SWD, PCBA Bring-up
- **Mechanical Design:** Fusion 360, SolidWorks, PID Control, ROS2

EDUCATION

[King Mongkut's University of Technology North Bangkok](#)

Jul. 2022 - Apr. 2026

Bachelor of Engineering in Robotics and Automation Systems

Bangkok, Thailand

GPA: 3.83 / 4.0 (First Class Honours) | TOEIC: 950 / 990 | Sakura Science Program (Japan) 1st Prize Winner